03. Exploratory Data Analysis

Use EDA to expose skew, class imbalance, feature relationships and data limitations.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

EDA present in the notebook

- Loads Malnutrition data.csv into pandas.
- Displays a sample of rows and inspects dataframe structure with info().
- Plots histograms for numeric/integer columns.
- Contains commented-out boxplot code for feature distributions by Status.
- In the oversampling notebook, examines Weight, Height and Age distributions/outliers using boxplots, histograms and scatterplots after resampling.

Main EDA findings

- No null values were present in the supplied CSV.
- Status is heavily concentrated in Stunting (696/837).
- Wasting has only four original examples, so estimates for that class are extremely uncertain.
- The archive does not document measurement units, sampling design, age range or the derivation of the income indicators.

Recommended EDA sequence

- Check duplicate rows and duplicate child identifiers if available.
- Validate plausible ranges and units for Age, Height and Weight; investigate impossible or inconsistent measurements with domain guidance.
- Plot class-conditional distributions for Height, Weight and Age; inspect overlap rather than assuming one feature determines a label.
- Inspect correlations and multicollinearity among input variables; verify the income flags are mutually exclusive and correctly encoded.
- Use scatterplots such as Height versus Weight, faceted by Status and Age; a single global scatterplot can hide age effects.
- Check whether class balance varies by age, sex or income group.

Avoid causal claims

A relationship between income indicators and Status does not establish that income caused a nutritional outcome. The dataset is small and observational; confounding, selection effects and label-definition issues may be substantial.